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Green Metals

Battery Metals Watch: The bear market is far from over

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Unresolved gluts. Despite the significant downside in battery metals prices, with nickel, lithium and cobalt prices down 60%, 80% and 65% from cycle peak, respectively, we believe it is too early to call a decisive end to these respective bear markets. Margin pressures have generated a meaningful degree of supply rationing effects over the past quarter, with incremental supply cuts in lithium (153kt, 10% global demand) and nickel (82kt, 2% global demand). Yet still significant supply pipelines combined with Western EV related demand downgrades, mean that the projected 2024 surpluses for lithium (GSe 150kt, 11% global demand) and nickel (GSe 224kt, 6% global demand) remain sizeable. Our expectation for cobalt's 2024 surplus is unchanged (GSe 15kt, 7% global demand), though the 2025 surplus has doubled in size (GSe 22kt, 9% global demand) on DRC supply upgrades. Recent short covering rallies in both lithium and nickel should therefore not be misconstrued as the end of the bear market, in our view. On a 12M basis we target 12%, 15%, 25% downside in cobalt ($26,000/t vs. prev. $28,000/t), nickel ($15,000/t) and lithium carbonate ($10,000/t vs. prev. $11,000/t) respectively.

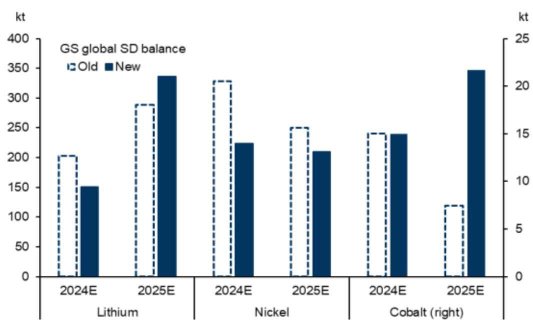
Western EV headwinds. Whilst rationing supply growth has been the key focus on price path, it is important to recognise that Western EV-tied demand downgrades are also working to partially offset supply effects. Continued price competition has contributed to profit pressure for Western OEM's, which has slowed the pace of new EV investment and penetration path. Indeed our US and European auto analysts lowered respective EV sales penetration forecasts by 5% and 2% for 2024 and an average of 11% and 1% over the proceeding 3 years. China EV penetration assumptions have proven more resilient, and therefore our expectations for global EV sales for this year have only fallen modestly (+19% y/y vs. prev +20%). Nonetheless, combined with slower battery capacity additions in China this year (c.400GWh in 2024E vs 460GWh in 2023), we expect the battery metals stocking impulse to be more modest this year. The potential for additional China EV stimulus presents the most obvious upside risk to this demand outlook, with signaling of a 'cash for clunkers' style stimulus a focus in this regard.

Lithium's prolonged trough. The recent short covering rally in lithium prices, with the onshore front-end carbonate contract rallying 25% over the past week, should not be interpreted as the end of the bear market, in our view. Since our previous update in mid-Q4, we have cut nearly 10% (153kt LCE) from our pre-disruption adjusted global lithium supply forecasts for 2024. Yet this largely reflects a trend of delays to ramp up path, meaning little to no impact on the supply path post-2024. Moreover, adjusting for disruption allowance and demand downgrades, our latest balance shows a still sizeable 150kt surplus in 2024 (vs. prev 202kt surplus) and 336kt surplus in 2025 (vs. prev 288kt surplus). Further supply rationing is needed to reduce both the 2024E surplus and now larger surplus in 2025E. The top end of the integrated cash cost curve is dominated by Chinese lepidolite ($8k-12k/t LCE) and integrated production using African concentrates ($7k-$13k/t LCE). In that context, we now expect China lithium carbonate pricing to average $11,100/t this year (vs $13,376/t) and cut our 12M target to $10,000/t (vs $11,000/t previously).

The bear market is far from over

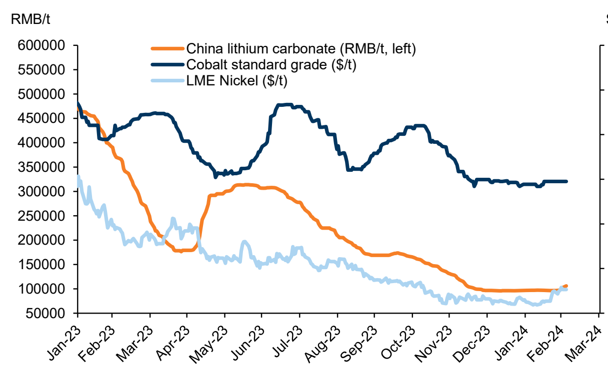
1. Unresolved gluts. Despite the significant downside in battery metals prices, with nickel, lithium and cobalt prices down 60%, 80% and 65% from cycle peak, respectively, we believe it is too early to call a decisive end to these respective bear markets. Margin pressures have generated a meaningful degree of supply rationing effects over the past quarter, with incremental supply cuts in lithium (153kt, 10% global demand) and nickel (82kt, 2% global demand). Yet still significant supply pipelines combined with Western EV related demand downgrades, mean that the projected 2024 surpluses for lithium (GSe 150kt, 11% global demand) and nickel (GSe 224kt, 6% global demand) remain sizeable. Our expectation for cobalt's 2024 surplus is unchanged (GSe 15kt, 7% global demand), though the 2025 surplus has doubled in size (GSe 22kt, 9% global demand) on DRC supply upgrades. Recent short covering rallies in both lithium and nickel should therefore not be misconstrued as the end of the bear market, in our view. On a 12M basis we target 12%, 15%, 25% downside in cobalt ($26,000/t vs. prev. $28,000/t), nickel ($15,000/t) and lithium carbonate ($10,000/t vs. prev. $11,000/t) respectively.

Exhibit 1: Sizeable surplus path persists for all the battery metals even with recent supply rationing effects



Source: Goldman Sachs Global Investment Research

Exhibit 2: Short covering moves in nickel and lithium should not be misconstrued as the end of bear markets

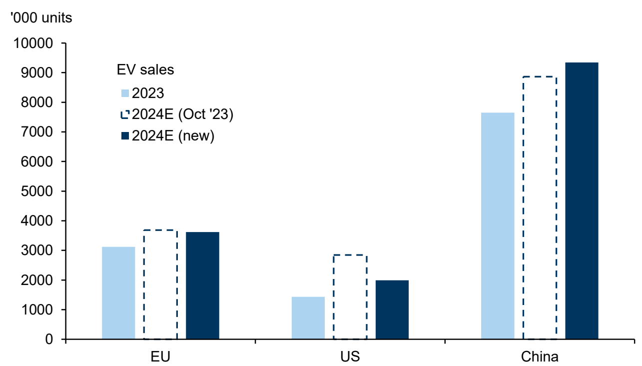


Source: Goldman Sachs Global Investment Research, Fastmarkets, SMM

2. Western EV slowdown. In the US, our auto analysts expect more difficult market conditions for EVs in the year ahead. A challenging pricing backdrop and delaying OEM EV ramp plans will likely lead to slower EV growth in 2024 (+554k units vs +1.2Mil units previously). On a similar note, our Europe auto team downwardly revised their EV penetration forecasts, mainly due to the withdrawal of subsidies and a shift from demand- to supply-driven EV adoption trends. Despite the decelerating penetration growth, EVs are still positioned to turn into the dominant technology in Europe in coming years. Even with the downgrade, we expect European EV sales to rise to 3.6Mil units (vs 3.1Mil units in 2023) as OEMs introduce several new models and ramp towards EU's 2025 CO2 targets. In China, OEM's continued price competition and new model launches should drive volume growth for EVs in 2024 and our auto analysts expect a demand pick-up in March, following more subdued February demand dragged down by CNY seasonality. In that context, China's passenger retails sales are estimated to rise to 9.3Mil units (vs 7.7Mil units in 2023). Globally, we expect EV sales

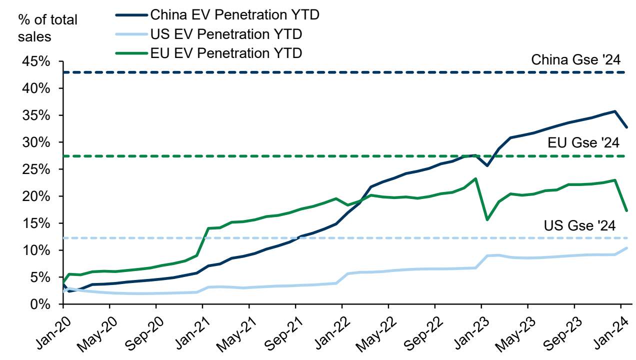
(PHEV+BEV) of 16.9Mil units in 2024 (+3.1Mil units y/y) compared to 13.8Mil units in 2023 (+3.6 Mil units y/y) based on our auto analysts' projections.

Exhibit 3: Western EV downgrades only partially offset by China upgrade



Source: Goldman Sachs Global Investment Research

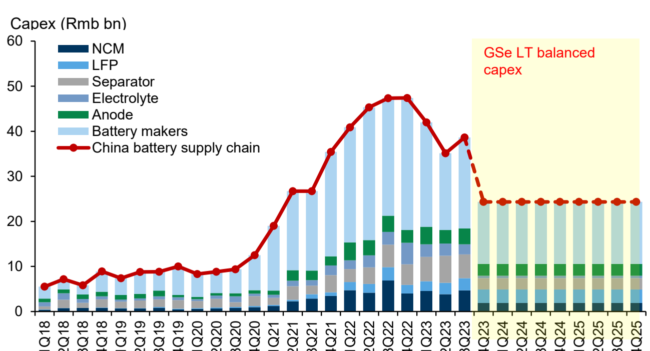
Exhibit 4: EV penetration rates are currently running below GS forecast for 2024



Source: Goldman Sachs Global Investment Research, Wind, IHS, Haver Analytics

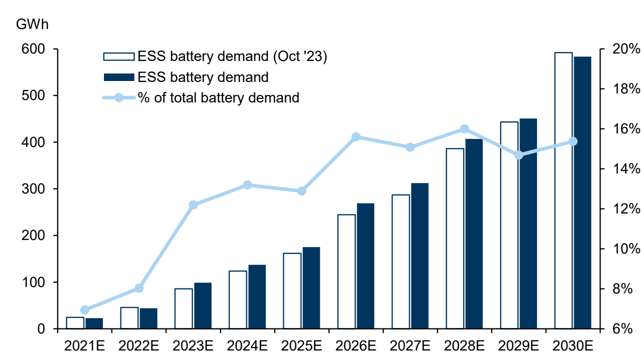
3. Nearer to the end of China's battery surplus. The softened EU and US EV penetration outlook has in turn lowered expectations on the global battery demand growth rate for this year, now expected to rise by 29% y/y from 35% previously (vs 31% y/y in 2023)**.** Notwithstanding the weaker demand path going forward, our battery analysts expect the Ex-China battery market to remain tight through 2025 due to delays in new battery capacity expansion. In China, the accelerated supply expansion and battery capex surge last year had pushed the battery balance into a surplus, weighing in turn on restocking demand for lithium. This year, GS expects the onshore battery surplus to peak in the first quarter, due to continued deceleration in battery sector capex (Exhibit 5) and a shift in the government policy stance, which aims to rationalise capital investment and mitigate China battery overcapacity risks. As opposed to the softer EV battery demand path, we have marginally upgraded our energy storage related demand on the strength of renewable installations growth in China. We expect ESS battery demand to account for 13% of total global battery demand (vs 12% last year and just 7% in 2021) and 15% by 2030, based on our sector analysts' projections.

Exhibit 5: China battery capex is set to fall into mid-decade, restraining current oversupply challenge



Source: Goldman Sachs Global Investment Research, Refinitiv Eikon

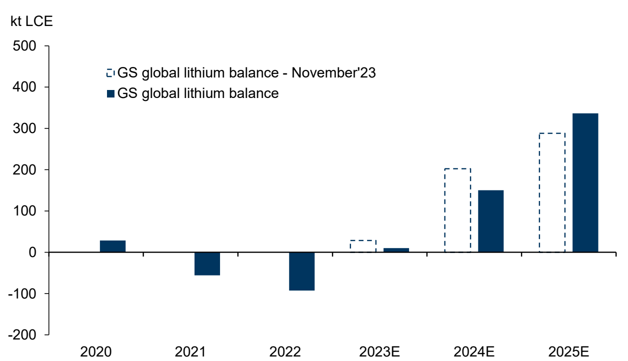
Exhibit 6: We expect ESS battery demand to rise on the strength of renewable installations growth in China



Source: Goldman Sachs Global Investment Research, BNEF, IEA

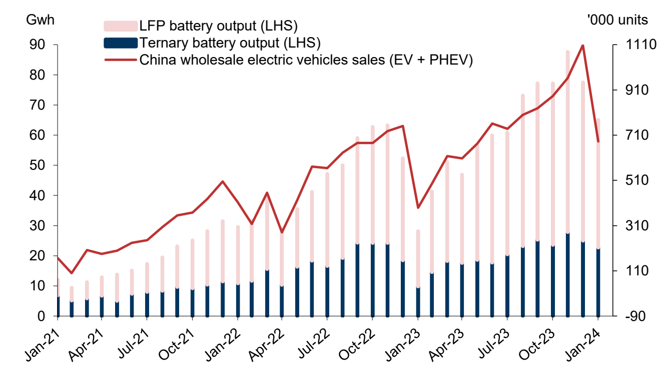
4. Lithium's prolonged trough. The recent short covering rally in lithium prices, with the onshore front-end carbonate contract rallying 25% over the past week, should not be interpreted as the end of the bear market, in our view. Since our previous update in mid-Q4, we have cut nearly 10% (153kt LCE) from our pre-disruption adjusted global lithium supply forecasts for 2024. Yet this largely reflects a trend of delays to ramp up path, meaning little to no impact on the supply path post-2024. Moreover, adjusting for disruption allowance and demand downgrades, our latest balance shows a still sizeable 150kt surplus in 2024 (vs. prev 202kt surplus) and 336kt surplus in 2025 (vs. prev 288kt surplus). Further supply rationing is needed to reduce both the 2024E surplus and now larger surplus in 2025E. The top end of the integrated cash cost curve is dominated by Chinese lepidolite ($8k-12k/t LCE) and integrated production using African concentrates ($7k-$13k/t LCE). In that context, we now expect China lithium carbonate pricing to average $11,100/t this year (vs $13,376/t) and cut our 12M target to $10,000/t (vs $11,000/t previously).

Exhibit 7: Prices are still substantially above the levels required to balance the market



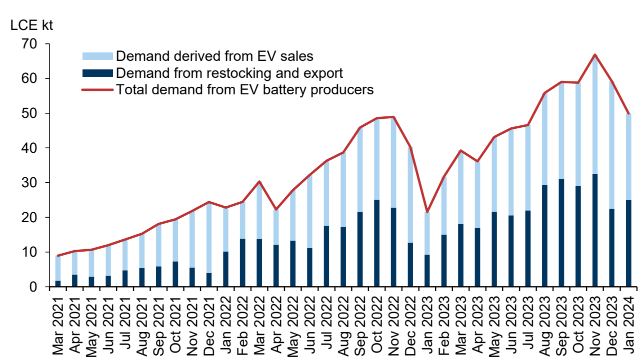
Source: Goldman Sachs Global Investment Research, Woodmac, SMM, Data compiled by Goldman Sachs Global Investment Research

Exhibit 8: Chinese EV sales and battery output have continued to increase, albeit at a slower pace of growth



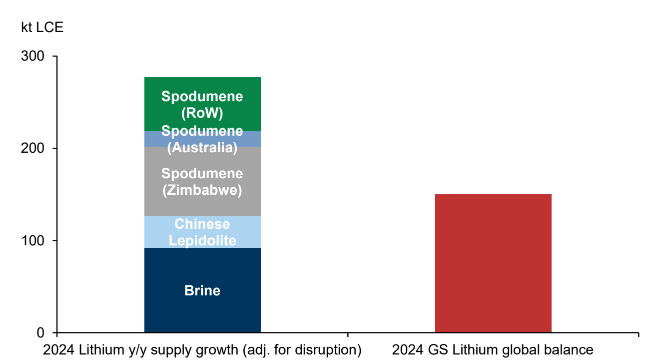
Source: Goldman Sachs Global Investment Research, CAAM, CPCA, Wind

Exhibit 9: Restocking demand is likely to remain modest in 2024



Source: Goldman Sachs Global Investment Research, Wind, SMM

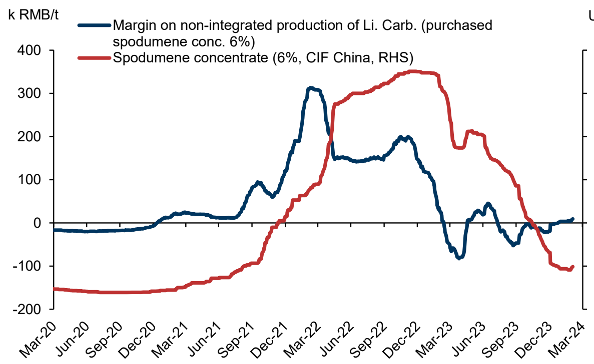
Exhibit 10: Lithium supply growth continues to outweigh surplus



*Lithium is in kt LCE

Source: Goldman Sachs Global Investment Research, Woodmac, BNEF

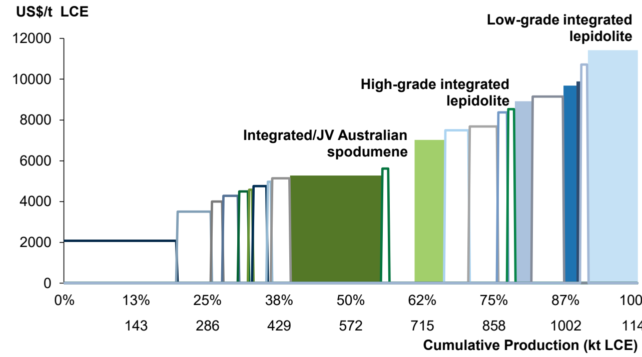
Exhibit 11: Downside in spodumene prices has supported a stabilisation in non-intergrated producer margins



Source: Goldman Sachs Global Investment Research, SMM

Exhibit 12: Lithium prices continue to remain at levels which are not generating material margin pressure on integrated producers

2024 cost curve, excluding royalties and VAT



Source: Goldman Sachs Global Investment Research, Data compiled by Goldman Sachs Global Investment Research, Woodmac

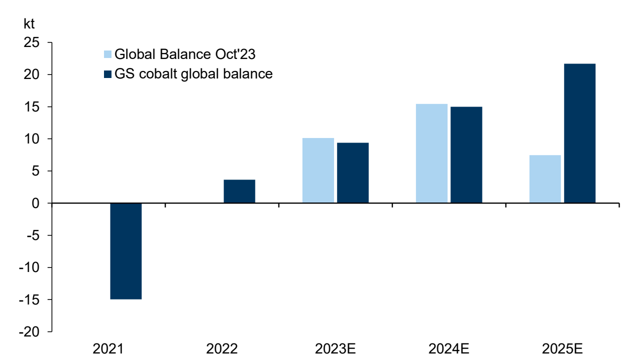
Exhibit 13: GS Lithium supply-demand balance

Li ('000 tonnes LCE)	2020	2021	2022E	2023E	2024E	2025E	2026E	2027E	2028E	2029E	2030E
Global demand											
Consumption - batteries	**200**	**361**	**562**	**786**	**989**	**1261**	**1547**	**1796**	**2152**	**2575**	**3090**
% change y/y	*42%*	*80%*	*56%*	*40%*	*26%*	*27%*	*23%*	*16%*	*20%*	*20%*	*20%*
EV	113	247	398	557	704	923	1115	1308	1560	1912	2339
ESS	10	17	33	73	101	129	195	214	266	278	327
E-buses, two-wheeler EVs	20	33	48	64	84	102	120	146	187	235	263
Portable electronics	39	41	46	47	48	50	53	56	60	63	67
Other	18	24	37	46	51	57	64	73	79	87	95
Consumption - ex batteries	**130**	**137**	**141**	**144**	**148**	**151**	**155**	**158**	**162**	**165**	**169**
% change y/y	*-2%*	*5%*	*3%*	*2%*	*3%*	*2%*	*2%*	*2%*	*2%*	*2%*	*2%*
Ceramics	32	34	35	36	37	38	39	40	41	42	43
Glass-ceramics	27	28	29	30	31	31	32	33	34	35	36
Other	71	74	77	78	80	82	83	85	86	88	90
% of global demand		15%	18%	6%	3%	6%	6%	5%	4%	3%	2%
Global demand (excl. restocking)	**331**	**498**	**703**	**930**	**1137**	**1413**	**1702**	**1954**	**2314**	**2740**	**3259**
Global demand (incl. restocking)	**351**	**570**	**830**	**990**	**1169**	**1501**	**1796**	**2045**	**2404**	**2831**	**3339**
% change y/y	*23%*	*63%*	*46%*	*19%*	*18%*	*28%*	*20%*	*14%*	*18%*	*18%*	*18%*
Global Refined Supply	**375**	**509**	**743**	**1045**	**1442**	**2125**	**2524**	**2879**	**3101**	**3255**	**3324**
Brine	**172**	**228**	**318**	**372**	**505**	**704**	**838**	**974**	**1057**	**1128**	**1152**
China	36	60	79	114	135	229	271	307	314	320	327
Ex-China	137	168	239	257	370	475	568	667	744	807	825
Spodumene	**189**	**236**	**342**	**534**	**747**	**1081**	**1322**	**1523**	**1633**	**1689**	**1706**
China	10	7	14	30	59	86	127	183	190	196	203
Ex-China	179	229	328	504	688	996	1195	1340	1444	1493	1503
Other (Lepidolite or Clay)	**13**	**45**	**83**	**140**	**190**	**339**	**363**	**382**	**410**	**439**	**467**
China	13	45	83	140	190	339	363	382	409	435	462
Ex-China	0	0	0	0	0	0	0	0	2	3	5
World output	**375**	**509**	**743**	**1045**	**1442**	**2125**	**2524**	**2879**	**3101**	**3255**	**3324**
% change y/y		*36%*	*46%*	*41%*	*38%*	*47%*	*19%*	*14%*	*8%*	*5%*	*2%*
Total output (adj. for disruption)	**375**	**504**	**721**	**967**	**1269**	**1764**	**2095**	**2389**	**2574**	**2702**	**2759**
% change y/y		*35%*	*43%*	*34%*	*31%*	*39%*	*19%*	*14%*	*8%*	*5%*	*2%*
Battery Scrap Supply	**5**	**10**	**17**	**33**	**50**	**73**	**91**	**93**	**121**	**219**	**336**
% change y/y		*2%*	*2%*	*3%*	*4%*	*4%*	*4%*	*4%*	*5%*	*8%*	*12%*
Global Balance	**29**	**-56**	**-93**	**10**	**150**	**336**	**390**	**438**	**290**	**90**	**-244**
as % of global supply	8%	-11%	-13%	1%	12%	19%	19%	18%	11%	3%	-9%

Source: Goldman Sachs Global Investment Research, Company data, BNEF, IEA, Woodmac

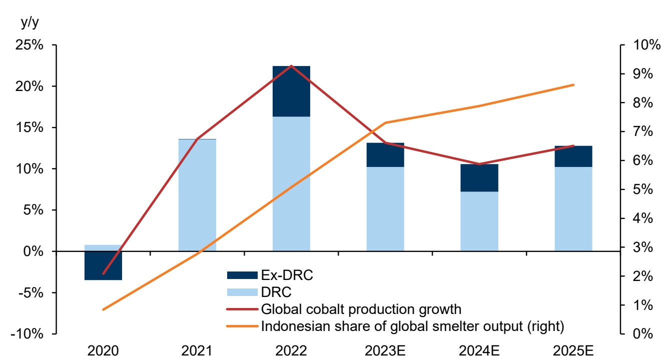
5. Cobalt's glut worsens. Cobalt prices have remained remarkably stable so far this year in comparison to the other battery metals. Standard grade cobalt has averaged just above $29,000/t, in a very tight range. We see this as unsustainable given the oversupply challenge in this market appears to increasing currently. Our projected 2024 surplus is unchanged (GSe 15kt, 7% global demand), but our 2025 surplus has doubled in size (GSe 22kt, 9% global demand). This ultimately reflects the influence of supply upgrades in the DRC focused on CMOC's Tenke Fungurume and Kisanfu operation (also DRC). Whilst a meaningful downgrade to the midterm production path at the Mutanda operation (Glencore, DRC) offers a tightening factor into the second half of the decade, we believe that will be too late to prevent large surpluses in the interim. As with previous cobalt bear markets, it is likely that some DRC producers will stockpile cobalt ore and only export copper concentrate, which may help to restrain the size of metals surplus albeit temporarily. However, it is also important to recognise the strong supply growth evident from the ramp up of new Indonesian intermediate and Chinese refining capacity. Indeed, China is now swinging into a net exporter status for cobalt metal, with evidence of Chinese cobalt brands now being offered actively into the European market. Unless the SRB steps up buying activity, the risk is this metal surplus generates persistent downside pressure on price. In this context, we now downgrade our average price forecasts for 2024 (GSe $27,5000/t vs. $30,000/t previously) and 2025 (GSe $25,000/t vs. $30,000/t previously).

Exhibit 14: Cobalt market remains structurally bearish on oversupply from DRC and Indonesia



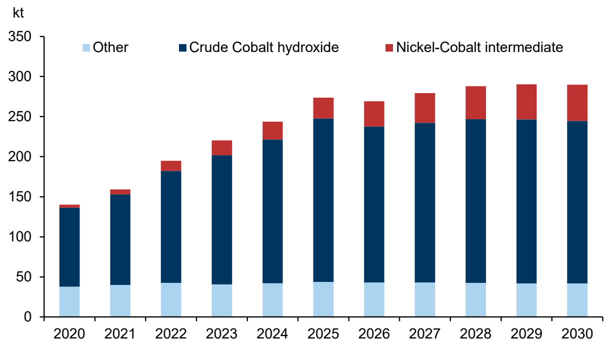
Source: Woodmac, BNEF, Fastmarkets, Goldman Sachs Global Investment Research

Exhibit 15: Rising MHP production in Indonesia should add metal volumes over 2024-25 as incremental Chinese refining capacity is brought online



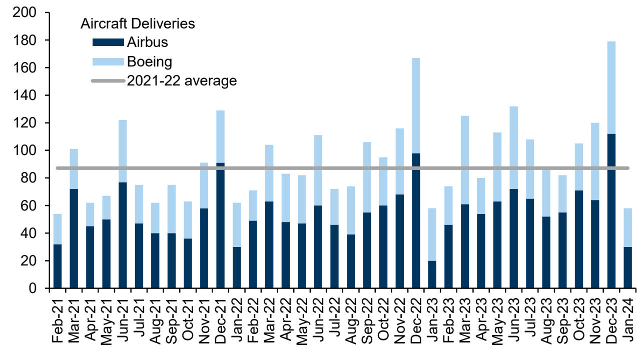
Source: Goldman Sachs Global Investment Research, Company data, SMM

Exhibit 16: Nickel-Co intermediate based production expected to be the main source of supply growth from 2026



Source: Goldman Sachs Global Investment Research, Woodmac

Exhibit 17: Non-battery cobalt demand has been a relative bright spot, particularly aerospace and defence industry



Source: Goldman Sachs Global Investment Research, Airbus, Boeing, Cirium

Exhibit 18: GS Cobalt supply-demand balance

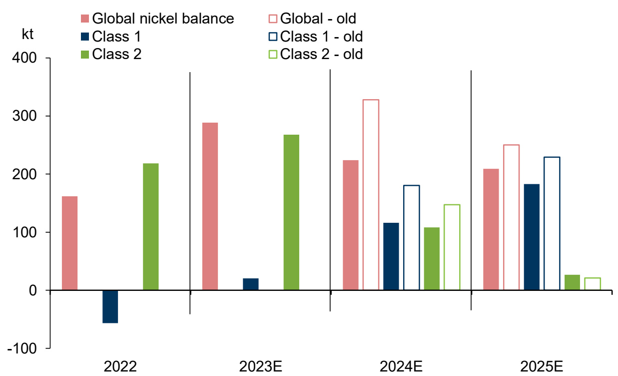
('000 tonnes)	2020	2021	2022E	2023E	2024E	2025E	2026E	2027E	2028E	2029E	2030E
Global demand											
Consumption - batteries	**84**	**112**	**119**	**138**	**151**	**172**	**192**	**209**	**226**	**260**	**289**
% of total	**59%**	**65%**	**65%**	**68%**	**69%**	**70%**	**72%**	**73%**	**74%**	**77%**	**78%**
% change y/y	*5%*	*33%*	*6%*	*16%*	*9%*	*14%*	*11%*	*9%*	*8%*	*15%*	*11%*
EV	22	46	54	69	78	96	109	122	131	159	182
ESS	2	2	3	6	7	8	12	12	15	16	17
E-buses, two-wheeler EVs	2	3	2	3	5	5	6	7	8	8	10
Portable electronics	55	57	55	56	57	58	60	63	67	71	75
Other	3	4	4	4	4	4	4	5	5	6	6
Consumption - ex batteries	**57**	**61**	**64**	**65**	**69**	**72**	**74**	**76**	**78**	**79**	**81**
% change y/y	*-7%*	*6%*	*4%*	*3%*	*5%*	*5%*	*3%*	*3%*	*2%*	*2%*	*2%*
Superalloys	17	17	19	20	21	22	23	23	24	24	25
Tool materials	11	12	12	13	13	13	14	14	14	14	14
Other	30	32	33	32	34	37	38	39	40	41	42
Global consumption	**142**	**173**	**182**	**203**	**220**	**244**	**266**	**285**	**303**	**339**	**370**
% change y/y	0%	22%	5%	11%	8%	11%	9%	7%	6%	12%	9%
Global Production											
World smelter output	**140**	**159**	**195**	**220**	**244**	**275**	**283**	**303**	**319**	**328**	**331**
% change y/y	*-3%*	*14%*	*22%*	*13%*	*11%*	*13%*	*3%*	*6.9%*	*5.6%*	*2.7%*	*0.9%*
DRC	96	115	141	161	177	202	200	208	217	220	219
% change y/y	*1%*	*20%*	*22%*	*14%*	*10%*	*14%*	*-1%*	*4%*	*4%*	*1%*	*0%*
Ex- DRC	44	44	54	59	67	73	83	95	103	108	112
% change y/y	*-10%*	*0%*	*22%*	*11%*	*12%*	*9%*	*14%*	*14%*	*8%*	*6%*	*4%*
World smelter output base	**140**	**159**	**195**	**220**	**244**	**274**	**269**	**279**	**288**	**290**	**290**
% change y/y	*-3%*	*14%*	*22%*	*13%*	*11%*	*12%*	*-2%*	*4%*	*3%*	*1%*	*0%*
Crude Co hydroxide	99	113	140	161	180	204	195	199	204	205	203
Other	41	46	55	59	64	70	74	80	84	86	87
Probable projects		**0**	**0**	**0**	**0**	**1**	**14**	**23**	**31**	**38**	**41**
Crude Co hydroxide		0	0	0	0	0	8	11	14	15	16
Other		0	0	0	0	1	6	13	18	22	25
World refined output	**135**	**153**	**188**	**213**	**235**	**265**	**273**	**292**	**308**	**316**	**319**
World output (adj. for disruption)	**135**	**153**	**180**	**205**	**226**	**255**	**262**	**283**	**299**	**307**	**309**
% change y/y		*14%*	*18%*	*14%*	*10%*	*13%*	*3%*	*8%*	*6%*	*3%*	*1%*
Battery Scrap Supply	**4**	**5**	**5**	**7**	**9**	**11**	**13**	**14**	**15**	**23**	**29**
% change y/y		*3%*	*3%*	*3%*	*4%*	*4%*	*5%*	*5%*	*5%*	*8%*	*9%*
Global Balance	**-3**	**-15**	**4**	**9**	**15**	**22**	**9**	**11**	**10**	**-9**	**-32**
as % of global supply	-2%	-10%	2%	5%	7%	9%	4%	4%	3%	-3%	-10%
Cash Prices (annual average)											
Current dollar ($/t)	33947	51119	69100	34790	27500	25000	32000	44000	44000	44000	44000
Current dollar ($/lb)	15.4	23.2	31.3	15.8	12.5	11.3	14.5	20.0	20.0	20.0	20.0

Prices refer to Fastmarkets cobalt metal standard grade

Source: Goldman Sachs Global Investment Research, Woodmac, BNEF

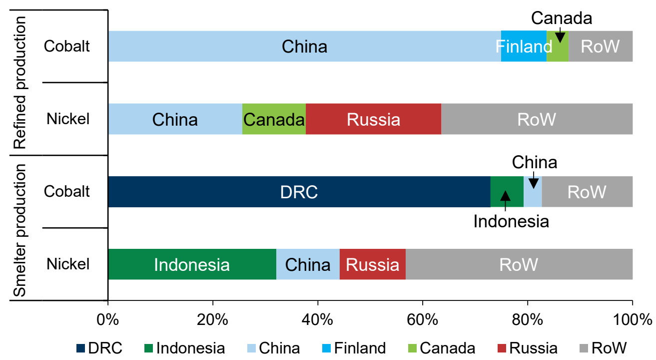
5. Nickel's brief reprieve. Similar to lithium, the nickel market has seen a phase of price-related supply rationing since our last update. Indeed, since November last year, we have cut our global primary production estimates by 80kt (2% global supply). This reflects cuts announced by a number of major ex-China nickel producers. Whilst this has reduced our projected 2024 surplus over the same period from 328kt (8% global demand) to 224kt (6% global demand), this still remains the third largest surplus faced by the nickel market in the last decade. Moreover, our balances still point to an average 240kty surplus over the 2025-27 period. Whilst high-cost producers are coming under pressure from falling prices, most of the supply growth is coming from low-cost producers in Indonesia and China. The ramp-up of chemical production through NPI to matte conversion and HPAL (GEM, CNGR, Jinchuan and Huayou Cobalt), alongside new metal capacities in China and Indonesia represent more than 75% of total supply growth this year. We expect Indonesian and Chinese refined assets to add c.180kty of supply in the form of metal and nickel sulphate in 2023-26. We note that the history of the nickel market suggests a substantial and sustained price overshoot into the cost curve is likely necessary to materially rebalance the market. We continue to target LME nickel at $15,000/t on a 12m horizon (average $16,000/t this year), which we estimate would move around 190kt of class 1 supply into loss-making territory. The continued scale of mid-term surplus projections suggests this degree of rationing is required to sustainably rebalance the market.

Exhibit 19: Nickel market to remain in a surplus into 2024 as supply continues to grow



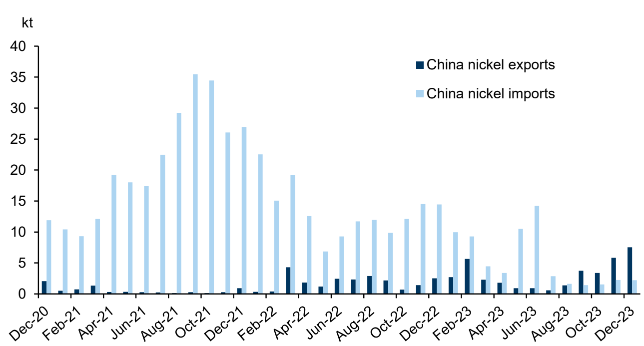
Source: Goldman Sachs Global Investment Research, INSG

Exhibit 20: Both class 1 nickel and cobalt prices are exposed to supply shock volatility due to high concentration
Based on 2023 GS estimates



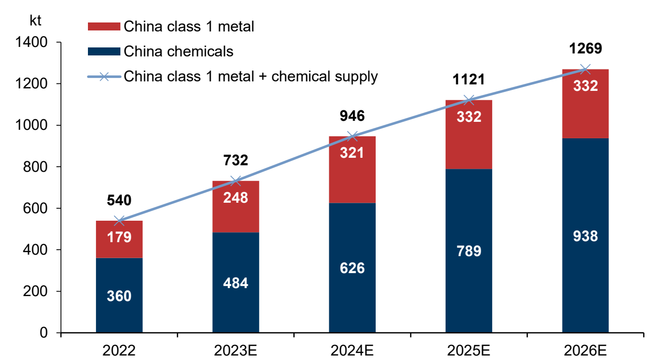
Source: Goldman Sachs Global Investment Research, Woodmac

Exhibit 21: China's swing into net refined nickel export status continues on refined capacity additions



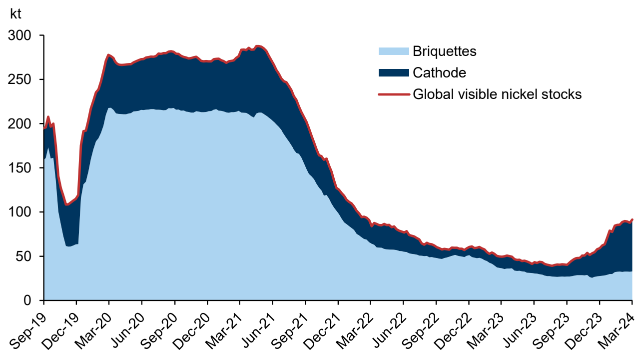
Source: Goldman Sachs Global Investment Research, SMM

Exhibit 22: Chinese refined assets should add c.180kty of metal and nickel sulphate supply each year over 2023-26E



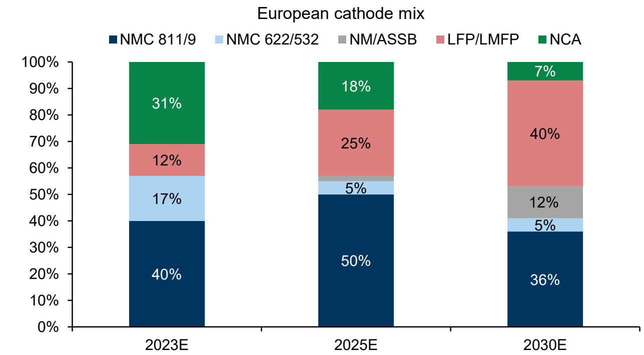
Source: Goldman Sachs Global Investment Research, Woodmac, Bloomberg, Company data

Exhibit 23: Evidence of the Class 1 nickel surplus is becoming more visible in exchange stocks levels



Source: Goldman Sachs Global Investment Research, Bloomberg, SMM

Exhibit 24: Rising share of LFP/LMFP batteries is weighing on long-term EV nickel demand



Source: Goldman Sachs Global Investment Research

Exhibit 25: GS Global Nickel SD Model

('000 tonnes)	2020	2021	2022	2023E	2024E	2025E	2026E	2027E	2028E	2029E	2030E
Consumption - DM											
US	126	128	136	148	166	212	289	334	370	412	462
% change y/y	-13%	2%	6%	9%	12%	28%	36%	16%	11%	11%	12%
Europe	292	345	319	328	356	392	425	459	501	553	603
% change y/y	-9%	18%	-8%	3%	8%	10%	9%	8%	9%	10%	9%
Japan	138	160	155	145	157	164	168	176	187	201	210
% change y/y	-10%	16%	-3%	-6%	8%	4%	3%	4%	7%	8%	4%
Other DM	7	8	9	9	9	9	9	9	9	10	10
% change y/y	-11%	7%	8%	2%	2%	2%	2%	2%	2%	2%	2%
Sub- DM	563	641	618	630	687	777	892	978	1067	1176	1285
% change y/y	-9%	14%	-4%	2%	9%	13%	15%	10%	9%	10%	9%
Consumption - EM											
China	1482	1539	1673	1890	2083	2268	2372	2466	2545	2703	2839
Indonesia	206	383	369	340	438	494	523	543	559	564	568
China & Indonesia	1688	1923	2042	2231	2522	2762	2896	3009	3104	3267	3406
% change y/y	10%	14%	6%	9%	13%	10%	5%	4%	3%	5%	4%
Other EM	263	311	234	243	261	275	283	285	298	317	338
% change y/y	-18%	18%	-25%	4%	7%	5%	3%	1%	4%	6%	7%
Sub- EM	1951	2234	2275	2474	2782	3037	3179	3294	3402	3583	3744
% change y/y	5%	15%	2%	9%	12%	9%	5%	4%	3%	5%	4%
Global Consumption	**2514**	**2875**	**2894**	**3105**	**3469**	**3814**	**4070**	**4272**	**4469**	**4759**	**5030**
% change y/y	1%	14%	1%	7%	12%	10%	7%	5%	5%	6%	6%
World ex-China & Indonesia	826	952	852	874	948	1052	1175	1263	1365	1492	1623
% change y/y	-12%	15%	-11%	3%	8%	11%	12%	8%	8%	9%	9%
Global Production											
Global Primary Production	**2478**	**2629**	**3046**	**3374**	**3665**	**3979**	**4241**	**4491**	**4601**	**4681**	**4711**
% change y/y	*7%*	*6%*	*16%*	*11%*	*9%*	*9%*	*7%*	*6%*	*2%*	*2%*	*1%*
China	766	739	883	1047	1186	1336	1465	1580	1631	1683	1711
% change y/y	*-4%*	*-4%*	*20%*	*19%*	*13%*	*13%*	*10%*	*8%*	*3%*	*3%*	*2%*
Indonesia	639	871	1143	1349	1462	1592	1680	1787	1829	1831	1831
% change y/y	*65%*	*36%*	*31%*	*18%*	*8%*	*9%*	*6%*	*6%*	*2%*	*0%*	*0%*
ex-China & Indonesia	1073	1019	1020	978	1016	1051	1097	1124	1141	1166	1168
% change y/y	*-4%*	*-5%*	*0%*	*-4%*	*4%*	*3%*	*4%*	*2%*	*2%*	*2%*	*0%*
Global Battery Scrap Supply	3	7	9	19	28	44	57	59	80	151	198
Global Balance	**-32**	**-239**	**162**	**289**	**224**	**209**	**228**	**277**	**211**	**73**	**-121**
Class 1 Nickel balance	-87	-233	-57	21	116	183	200	216	132	55	-89
Class 2 Nickel balance	54	-5	218	268	108	27	28	62	79	18	-32
Cash Prices (annual average)											
Current dollars ($/t)	13803	18474	28316	21526	16000	15500	-	-	-	-	-
Current dollars (c/lb)	626	838	1284	976	726	703	-	-	-	-	-

Source: Goldman Sachs Global Investment Research, Woodmac, SMM, BNEF

Appendix

Exhibit 26: GS Cobalt Green Demand Model

GS Cobalt Green Demand Model										
000 tonnes	2021	2022E	2023E	2024E	2025E	2026E	2027E	2028E	2029E	2030E
Europe	**9**	**7**	**8**	**9**	**12**	**16**	**20**	**24**	**31**	**37**
EV	8	7	8	8	12	15	18	21	28	34
ESS	0	0	1	1	1	2	2	2	3	3
YoY growth	*82%*	*-21%*	*15%*	*12%*	*37%*	*30%*	*23%*	*20%*	*30%*	*19%*
US	**4**	**4**	**6**	**8**	**14**	**25**	**31**	**33**	**39**	**44**
EV	4	4	5	7	13	23	28	30	35	41
ESS	0	0	0	1	1	2	3	3	4	4
YoY growth	*38%*	*-8%*	*47%*	*38%*	*80%*	*78%*	*23%*	*7%*	*17%*	*14%*
China	**31**	**43**	**57**	**63**	**72**	**74**	**76**	**79**	**92**	**101**
EV	30	40	52	58	66	67	69	71	84	92
ESS	2	2	5	5	6	7	7	8	8	9
YoY growth	*145%*	*37%*	*33%*	*11%*	*14%*	*3%*	*3%*	*4%*	*16%*	*10%*
RoW	**4**	**3**	**4**	**5**	**6**	**6**	**7**	**10**	**13**	**16**
EV	4	3	4	5	5	5	6	9	12	15
ESS	0	0	0	0	0	1	1	1	1	1
YoY growth	*39%*	*-20%*	*25%*	*24%*	*10%*	*-3%*	*27%*	*36%*	*40%*	*21%*
Global cobalt green demand	**51**	**59**	**78**	**90**	**109**	**127**	**141**	**154**	**184**	**208**
EV	46	54	69	78	96	109	122	131	159	182
ESS	2	3	6	7	8	12	12	15	16	17
E-buses, two-wheeler EVs	3	2	3	5	5	6	7	8	8	10
YoY growth	*100%*	*15%*	*32%*	*16%*	*22%*	*16%*	*11%*	*9%*	*20%*	*13%*

Source: Goldman Sachs Global Investment Research

Exhibit 27: GS Lithium Green Demand Model

GS Lithium Green Demand Model										
'000 tonnes LCE	2021	2022E	2023E	2024E	2025E	2026E	2027E	2028E	2029E	2030E
Europe	**49**	**54**	**69**	**86**	**126**	**177**	**227**	**299**	**389**	**496**
EV	46	49	61	76	111	150	195	256	339	435
ESS	3	4	8	11	15	26	32	44	49	61
YoY growth	*90%*	*10%*	*29%*	*26%*	*46%*	*40%*	*28%*	*32%*	*30%*	*28%*
US	**23**	**29**	**48**	**74**	**143**	**274**	**352**	**417**	**487**	**595**
EV	21	27	42	65	125	233	303	356	425	522
ESS	1	2	6	9	17	41	49	61	62	73
YoY growth	*44%*	*28%*	*64%*	*54%*	*93%*	*92%*	*29%*	*18%*	*17%*	*22%*
China	**169**	**322**	**478**	**596**	**725**	**800**	**863**	**989**	**1147**	**1357**
EV	158	298	423	521	636	681	742	845	1002	1191
ESS	11	24	55	75	89	119	121	144	146	167
YoY growth	*156%*	*90%*	*48%*	*25%*	*22%*	*10%*	*8%*	*15%*	*16%*	*18%*
RoW	**23**	**25**	**35**	**49**	**58**	**60**	**80**	**120**	**167**	**217**
EV	21	23	31	43	51	51	69	103	146	190
ESS	1	2	4	6	7	9	11	17	21	27
YoY growth	*45%*	*11%*	*39%*	*40%*	*18%*	*4%*	*33%*	*50%*	*39%*	*30%*
Global lithium green demand	**296**	**478**	**694**	**890**	**1154**	**1430**	**1668**	**2013**	**2425**	**2929**
EV	247	398	557	704	923	1115	1308	1560	1912	2339
ESS	17	33	73	101	129	195	214	266	278	327
E-buses, two-wheeler EVs	33	48	64	84	102	120	146	187	235	263
YoY growth	*107%*	*61%*	*45%*	*28%*	*30%*	*24%*	*17%*	*21%*	*20%*	*21%*

Source: Goldman Sachs Global Investment Research

Exhibit 28: GS Nickel Green Demand Model

GS Nickel Green Demand Model										
'000 tonnes	2021	2022E	2023E	2024E	2025E	2026E	2027E	2028E	2029E	2030E
Europe	**45**	**41**	**50**	**59**	**83**	**114**	**146**	**187**	**238**	**287**
EV	38	33	40	49	71	98	126	163	213	261
ESS	1	1	2	2	3	5	6	8	8	9
Wind	6	7	7	8	9	11	14	16	16	17
YoY growth	*65%*	*-9%*	*21%*	*19%*	*41%*	*37%*	*28%*	*28%*	*28%*	*21%*
US	**24**	**21**	**32**	**47**	**87**	**158**	**200**	**234**	**275**	**323**
EV	18	18	28	42	80	147	188	220	262	308
ESS	1	1	2	2	4	8	9	11	11	12
Wind	6	3	3	3	4	3	3	3	3	3
YoY growth	*20%*	*-11%*	*51%*	*46%*	*84%*	*81%*	*27%*	*17%*	*17%*	*18%*
China	**161**	**228**	**321**	**379**	**454**	**498**	**553**	**626**	**729**	**826**
EV	136	201	283	337	407	445	489	556	655	749
ESS	5	8	14	16	17	21	20	24	24	24
Wind	21	19	24	27	30	32	43	47	50	53
YoY growth	*90%*	*41%*	*41%*	*18%*	*20%*	*10%*	*11%*	*13%*	*16%*	*13%*
RoW	**26**	**23**	**30**	**36**	**41**	**42**	**54**	**75**	**103**	**127**
EV	18	15	20	27	31	32	44	64	90	113
ESS	1	1	1	1	1	2	2	3	4	4
Wind	8	7	9	8	8	8	8	8	8	10
YoY growth	*41%*	*-11%*	*28%*	*19%*	*15%*	*3%*	*27%*	*40%*	*37%*	*24%*
Global green demand nickel	**256**	**311**	**428**	**514**	**655**	**797**	**934**	**1100**	**1318**	**1532**
EV	189	235	327	401	520	645	754	898	1107	1301
ESS	8	11	19	22	25	35	37	45	47	50
E-buses, two-wheeler EVs	18	29	40	47	58	62	74	82	86	99
Wind	40	36	42	45	52	55	68	74	78	83
YoY growth	*70%*	*21%*	*38%*	*20%*	*27%*	*22%*	*17%*	*18%*	*20%*	*16%*

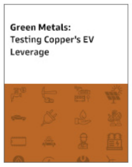
Source: Goldman Sachs Global Investment Research

Related Research

Green Metals: Battery Metals Watch: The rationing phase

26 Oct 2023

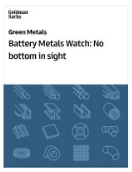
Too early to call the bottom. Despite the significant downside in battery metals prices this year, with nickel, lithium and cobalt prices down 60%, 70% and 60% from last year's peak, respectively, we believe it is too early to call an end to these respective bear markets.



Green Metals: Testing Copper's EV Leverage

5 Jul 2023

EVs already core to copper green demand surge. The electrification of the transportation sector - accounting for 22% of global emissions - is a key pillar of copper's bullish story. Due to copper's highly conductive and ductile nature, it is an ideal candidate for transformation and transmission of electrical energy in EVs.



Green Metals: Battery Metals Watch: No bottom in sight

7 Jun 2023

Structurally bearish fundamentals on oversupply. Battery metals prices have fallen significantly YTD, with China lithium carbonate prices declining 41%, nickel falling 29% and cobalt metal down 28%. Whilst this correction in prices has been predominantly tied to a sluggish start to the year in China's EV sector, we think that part of the sell-off also reflects respective markets' focus on strong supply trends across the complex.

Green Metals: Demand revolution gathers momentum

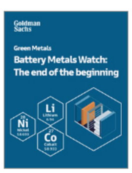
16 Mar 2023

While many investors can see the structural demand surge for green metals, we believe its true impact on fundamentals remains underappreciated in the market today. Green demand is at the heart of the metal supercycle—and the drivers are diversifying. The most important development over the last year has been the rise in strategic industrial competition between China, Europe, and the US as they race to attract clean tech capital amid the rise of programs like the US IRA and the European Green Deal Industrial Plan.

Green Metals: Battery Metals Watch: Lithium's delayed decline

9 Nov 2022

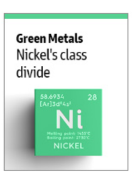
In Q2 this year, we called for peaking battery metals prices as mounting supply responses were set to catch up to the accelerating demand. Since then, the cobalt price has fallen 35% and nickel is down 40%, but onshore lithium prices have risen (24%) to record new levels. For lithium, whilst the supply response we had estimated has continued to gather pace, as evidenced by surging Chinese lithium carbonate output, LFP battery expansion related restocking demand and higher-than-expected EV sales in China due to the exceptional stimulus have kept the market tighter than expected.



Green Metals: Battery Metals Watch: The end of the beginning

29 May 2022

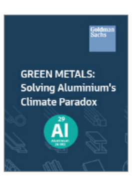
Battery metals – cobalt, lithium and nickel – will power the green industrial revolution, facing a wave of demand comparable to that of copper and iron ore during China's rapid growth in the 2000's. With climate change top of mind, investors are fully aware that battery metals will play a crucial role in the 21st century global economy, just as bulk and base metals did before them. Yet despite this exponential demand profile, we see the battery metals bull market as over for now.



Green Metals: Nickel's class divide

28 Apr 2022

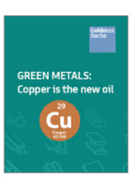
At the beginning of the year, nickel's place within green metals was as a key competitor in the race for mineral dominance of energy storage. Now, it sits at the intersection of Europe's push for decarbonisation and energy independence. At the heart of Europe's strategy lies its desire to rapidly electrify its transportation sector - a source of 20% of its emissions and c.2mb/d of Russian oil imports.



Green Metals: Solving Aluminium's Climate Paradox

20 Jun 2021

At the heart of the coming surge in green aluminium demand lies a paradox: aluminium is a key input required to produce decarbonising technologies like EV's and solar power, yet its own production is very carbon intensive, generating 2% of all global emissions. This paradox begs the question: how can we secure enough aluminium to effectively decarbonise, while keeping the climate impact of the path to net zero to a minimum? In our view, the resolution of this paradox will drive a structural bull market in aluminium over the next half decade, driven by the necessity to grow supply to meet green demand while cutting emissions to prevent a climate catastrophe.



Green Metals: Copper is the new oil

13 Apr 2021

The critical role copper will play in achieving the Paris climate goals cannot be overstated. As the most cost-effective conductive material, copper sits at the heart of capturing, storing and transporting these new sources of energy. Leveraging equity analysis across EVs, wind, solar, and battery technology, we estimate that by 2030, copper demand will grow nearly 600%. Yet the copper market as it currently stands is not prepared for this demand, and producers are leery of committing to new supply. We significantly increase our price forecasts and explore the implications across the metals and mining complex.

Disclosure Appendix

Reg AC

We, Nicholas Snowdon and Lavinia Forcellese, hereby certify that all of the views expressed in this report accurately reflect our personal views, which have not been influenced by considerations of the firm's business or client relationships.

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